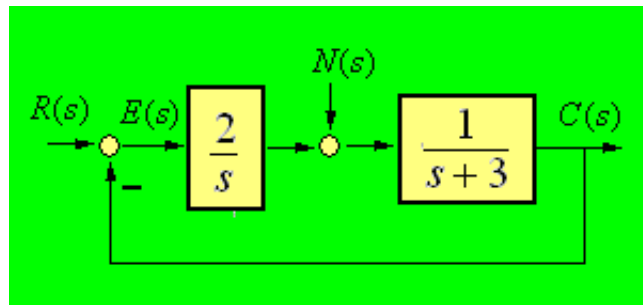


例 7 系统结构图如图所示

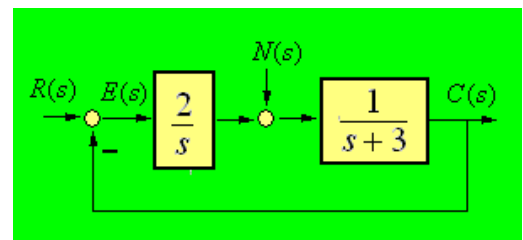


- (1) 求当输入 $r(t) = 1(t)$ 时系统的响应 $c_r(t)$;
- (2) 求当干扰 $n(t) = \delta(t)$ 时系统的响应 $c_n(t)$;
- (3) 求当初条件 $\begin{cases} c(0) = -1 \\ c'(0) = 0 \end{cases}$ 时系统的自由响应 $c_0(t)$;
- (4) 求在上述三个因素同时作用下系统的总输出 $c(t)$;
- (5) 求在上述三个因素同时作用下系统的总偏差 $e(t)$

(1) 求当输入 $r(t) = 1(t)$ 时系统的响应 $c_r(t)$;

解. 系统开环传递函数

$$G(s) = \frac{2}{s(s+3)} \quad K = \frac{2}{3}$$



系统闭环传递函数为

$$\Phi(s) = \frac{C(s)}{R(s)} = \frac{\frac{2}{s(s+3)}}{1 + \frac{2}{s(s+3)}} = \frac{2}{s(s+3) + 2} = \frac{2}{s^2 + 3s + 2} = \frac{2}{(s+1)(s+2)}$$

$$C_r(s) = \Phi(s)R(s) = \frac{2}{(s+1)(s+2)} \cdot \frac{1}{s} = \frac{2}{s(s+1)(s+2)}$$

$$= \frac{C_0}{s} + \frac{C_1}{s+1} + \frac{C_2}{s+2}$$

$$C_0 = \lim_{s \rightarrow 0} s \cdot C_r(s) = \lim_{s \rightarrow 0} \frac{2}{(s+1)(s+2)} = \frac{2}{2} = 1$$

$$C_1 = \lim_{s \rightarrow -1} (s+1) \cdot C_r(s) = \lim_{s \rightarrow -1} \frac{2}{s(s+2)} = \frac{2}{-1} = -2$$

$$C_2 = \lim_{s \rightarrow -2} (s+2) \cdot C_r(s) = \lim_{s \rightarrow -2} \frac{2}{s(s+1)} = \frac{2}{2} = 1$$

$$C_r(s) = \frac{1}{s} - \frac{2}{s+1} + \frac{1}{s+2}$$

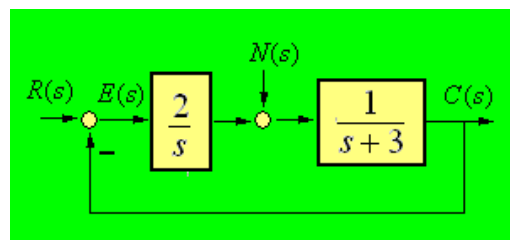
$$c_r(t) = 1 - 2e^{-t} + e^{-2t}$$

(2) 求当干扰 $n(t) = \delta(t)$ 时系统的响应 $c_n(t)$;

解. 干扰作用下系统的传递函数为

$$\Phi_n(s) = \frac{C(s)}{N(s)} = \frac{\frac{1}{s+3}}{1 + \frac{2}{s(s+3)}}$$

$$= \frac{s}{(s+1)(s+2)}$$



$$C_n(s) = \Phi_n(s)N(s) = \frac{s}{(s+1)(s+2)} \cdot 1$$

$$= \frac{s}{(s+1)(s+2)} = \frac{C_{n1}}{s+1} + \frac{C_{n2}}{s+2}$$

$$C_{n1} = \lim_{s \rightarrow -1} (s+1) \cdot C_n(s) = \lim_{s \rightarrow -1} \frac{s}{s+2} = \frac{-1}{1} = -1$$

$$C_{n2} = \lim_{s \rightarrow -2} (s+2) \cdot C_n(s) = \lim_{s \rightarrow -2} \frac{s}{s+1} = \frac{-2}{-1} = 2$$

$$C_n(s) = \frac{-1}{s+1} + \frac{2}{s+2}$$

$$c_n(t) = -e^{-t} + 2e^{-2t}$$

(3) 求当初条件 $\begin{cases} c(0) = -1 \\ c'(0) = 0 \end{cases}$ 时系统的自由响应 $c_0(t)$ 。

$$\Phi(s) = \frac{C(s)}{R(s)} = \frac{2}{s^2 + 3s + 2}$$

$$(s^2 + 3s + 2)C(s) = 2R(s)$$

$$c'' + 3c' + 2c = 2r$$

系统的齐次微分方程为

$$c'' + 3c' + 2c = 0$$

$$[s^2 C(s) - sc(0) - c'(0)] + 3[sC(s) - c(0)] + 2C(s) = 0$$

$$[s^2 + 3s + 2]C(s) + (s + 3) = 0$$

$$C_0(s) = \frac{-(s+3)}{s^2 + 3s + 2} = \frac{-(s+3)}{(s+1)(s+2)} = \frac{C_{01}}{s+1} + \frac{C_{02}}{s+2}$$

$$C_{01} = \lim_{s \rightarrow -1} (s+1) \cdot C_0(s) = \lim_{s \rightarrow -1} \frac{-(s+3)}{s+2} = -2$$

$$C_{02} = \lim_{s \rightarrow -2} (s+2) \cdot C_0(s) = \lim_{s \rightarrow -2} \frac{-(s+3)}{s+1} = 1$$

$$C_0(s) = \frac{-2}{s+1} + \frac{1}{s+2}$$

$$c_0(t) = -2e^{-t} + e^{-2t}$$

(4) 求在上述三个因素同时作用下系统的总输出 $c(t)$ 。

$$c_r(t) = 1 - 2e^{-t} + e^{-2t}$$

$$c_n(t) = -e^{-t} + 2e^{-2t}$$

$$c_0(t) = -2e^{-t} + e^{-2t}$$

$$c(t) = c_r(t) + c_n(t) + c_0(t) = 1 - 5e^{-t} + 4e^{-2t}$$

$$C_r(s) = \Phi(s)R(s) = \frac{2}{s^2 + 3s + 2} \cdot \frac{1}{s} = \frac{2}{s(s+1)(s+2)}$$

$$C_n(s) = \Phi_n(s)N(s) = \frac{s}{(s+1)(s+2)} \cdot 1 = \frac{s}{(s+1)(s+2)}$$

$$C_0(s) = \frac{-(s+3)}{s^2 + 3s + 2} = \frac{-(s+3)}{(s+1)(s+2)}$$

$$C_{r+n+0}(s) = C_r(s) + C_n(s) + C_0(s) = \frac{2 - s(s+3) + s^2}{s(s+1)(s+2)}$$

$$= \frac{-3s+2}{s(s+1)(s+2)} = \frac{C_0}{s} + \frac{C_1}{s+1} + \frac{C_2}{s+2}$$

$$C_0 = \lim_{s \rightarrow 0} \frac{-3s+2}{(s+1)(s+2)} = 1$$

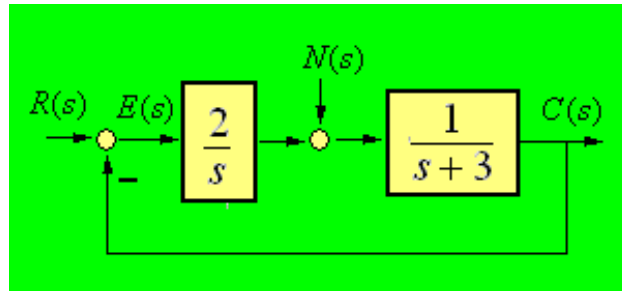
$$C_1 = \lim_{s \rightarrow -1} \frac{-3s+2}{s(s+2)} = -5$$

$$C_2 = \lim_{s \rightarrow -2} \frac{-3s+2}{s(s+1)} = 4$$

$$C_{r+0+n}(s) = \frac{1}{s} - \frac{5}{s+1} + \frac{4}{s+2}$$

$$c(t) = 1 - 5e^{-t} + 4e^{-2t}$$

(5) 求在上述三个因素同时作用下系统的总偏差 $e(t)$ 。



$$e(t) = r(t) - c(t) = 5e^{-t} - 4e^{-2t}$$